

Design and Development of a 6-DOF Robotic Arm Using Arduino

Rishika R-23BMH1050

Christa Daryn P-23BMH1126

Introduction

Robotic arms are widely used in industries for automation tasks such as assembly, pick-and-place operations, welding, and packaging. A robotic arm with multiple degrees of freedom (DOF) can mimic the movements of a human arm and perform complex manipulations.

This project focuses on the **design and development of a low-cost 6-DOF robotic arm prototype** using an **Arduino Uno microcontroller and servo motors**. The system demonstrates basic robotic manipulation and control, making it suitable for educational purposes and introductory robotics research.

Objective

The primary objectives of this project are:

- To design and construct a **6-degree-of-freedom robotic arm**.
- To control multiple joints using **servo motors (SG90 motor) and Arduino Uno**.
- To develop a **low-cost prototype for robotics learning and experimentation**.

System Overview

The robotic arm consists of **six rotational joints**, each driven by a servo motor, allowing the arm to move in multiple directions similar to a human arm.

The system includes:

- Arduino Uno microcontroller
- Multiple servo motors
- Robotic arm structure (made from wood/cardboard prototype)
- Electrical wiring and power supply
- Control program uploaded to Arduino

Each servo motor controls one degree of freedom of the arm, enabling precise angular positioning.

Hardware Components

Component	Description
Arduino Uno	Microcontroller used to control all servo motors
Servo Motors	Provide rotational movement for each joint
Structural Frame	Wooden/cardboard prototype forming the arm body
Jumper Wires	Used for electrical connections
Power Supply	Provides power to Arduino and servos

Degrees of Freedom (DOF)

The robotic arm is designed with **six degrees of freedom**, allowing flexible motion.

1. **Base Rotation** – Rotates the arm horizontally.
2. **Shoulder Joint** – Moves the arm up and down.
3. **Elbow Joint** – Provides bending motion similar to a human elbow.
4. **Wrist Pitch** – Allows vertical wrist movement.
5. **Wrist Rotation** – Rotates the end effector.
6. **Gripper Motion** – Opens and closes the gripper to pick objects.

These movements allow the robotic arm to reach different positions and manipulate objects.

Working Principle

The Arduino Uno controls the servo motors using **Pulse Width Modulation (PWM)** signals.

1. Commands are sent to Arduino.
2. Arduino generates PWM signals.
3. Each servo motor rotates to a specified angle.
4. The coordinated movement of all servos produces arm motion.

By controlling servo angles, the arm can perform **precise positioning and object manipulation**.

Design and Construction of Robotic Manipulator

1. 3D Design Phase

The robotic manipulator was initially designed using SolidWorks. The following steps were carried out during the design phase:

Design Steps:

- Created 3D models of base, links, joints, and gripper using SolidWorks
- Assigned material properties (Aluminium 6061-T6) to all components
- Calculated mass properties using SolidWorks mass analysis tool
- Analyzed weight distribution of the robotic arm
- Modified design to reduce overall weight and improve stability
- Finalized optimized lightweight design for fabrication

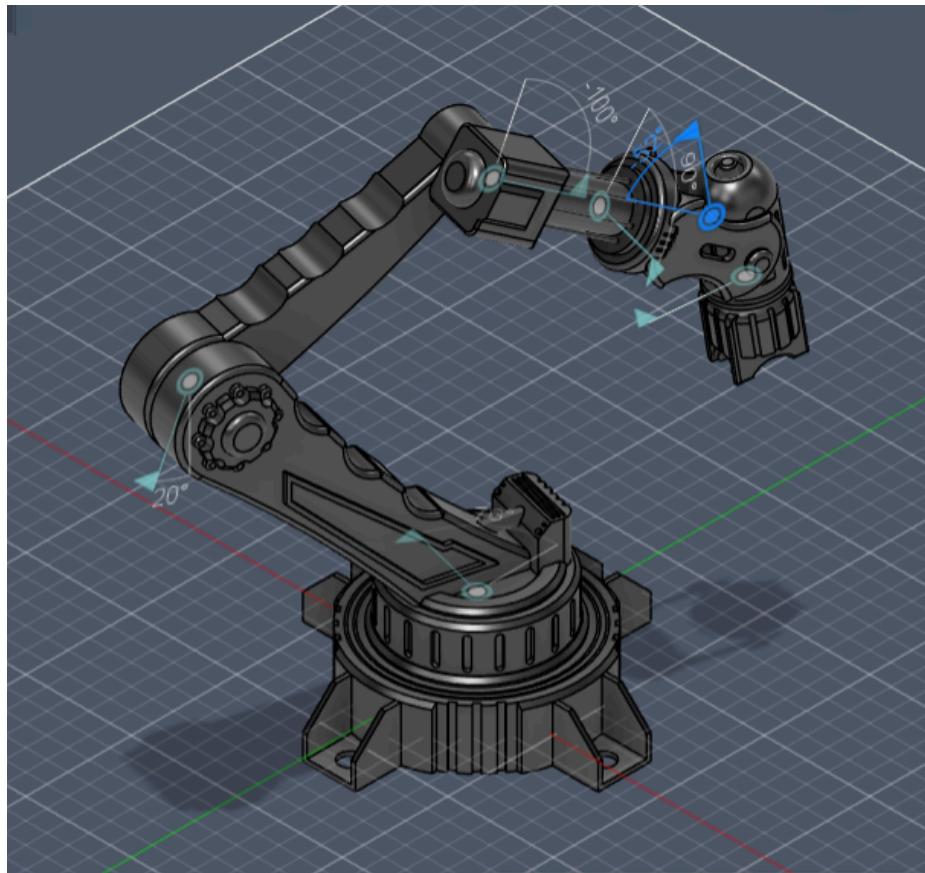


Figure 1: 6DOF robotic manipulator

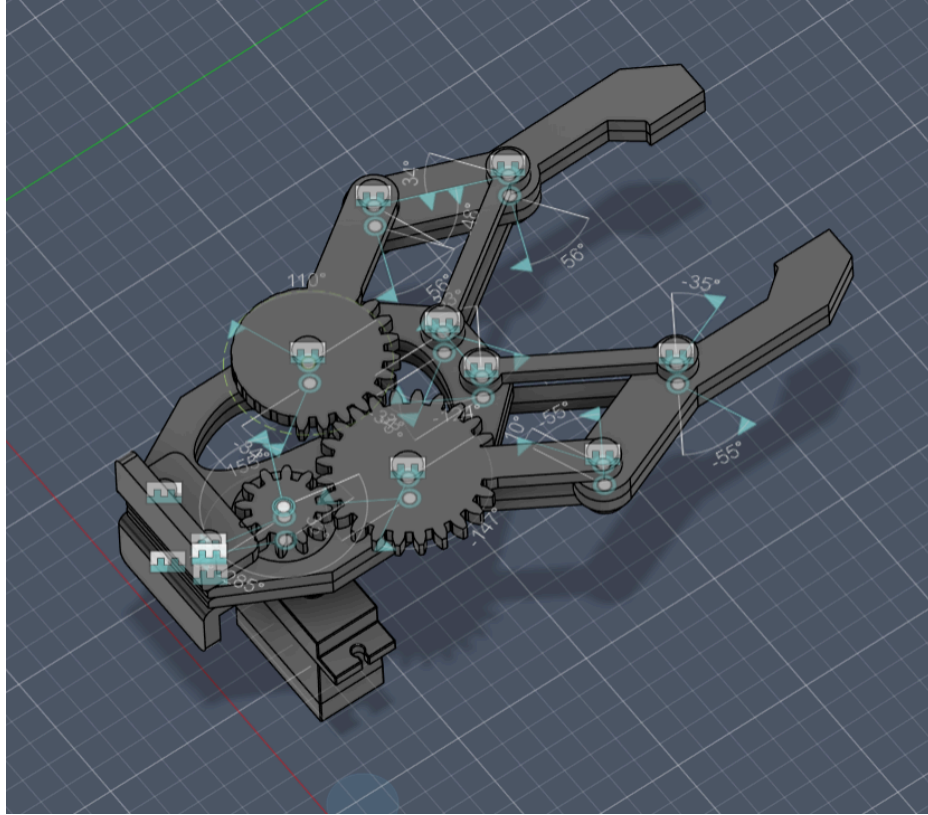


Figure 2: Gripper

2. Physical Construction Phase

After finalizing the 3D design, the robotic arm prototype was constructed using lightweight wooden/cardboard materials, making the structure easy to fabricate and modify.

Key Construction Features:

- Layered cardboard/wood joints for flexibility
- Servo motors embedded in joint housings
- Mechanical linkage for gripper control
- Circular base platform for arm rotation

The physical model was slightly modified from the initial 3D design based on weight calculations and structural requirements. The final design provided sufficient stability while maintaining a lightweight structure suitable for robotic operation.

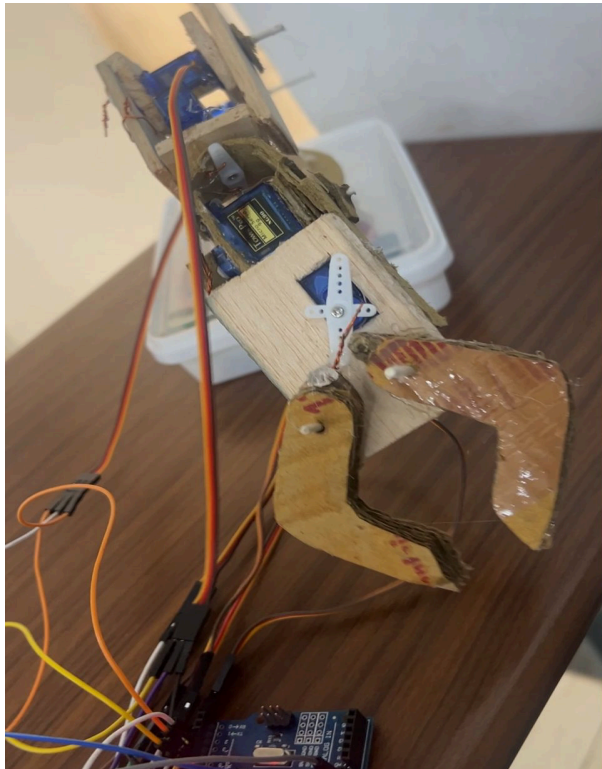
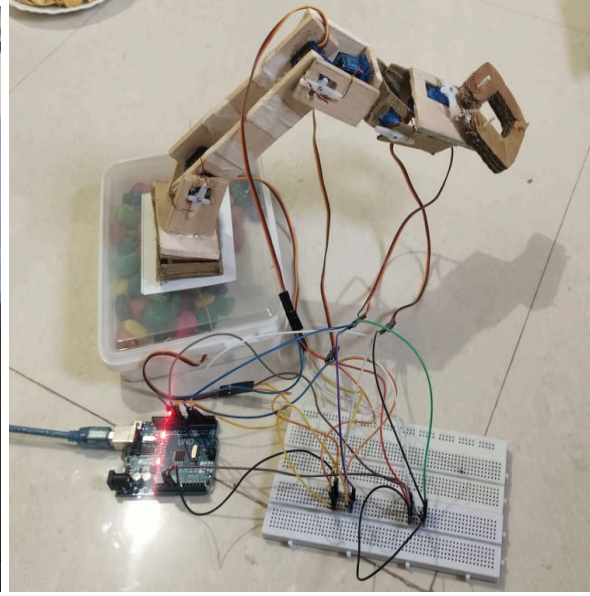
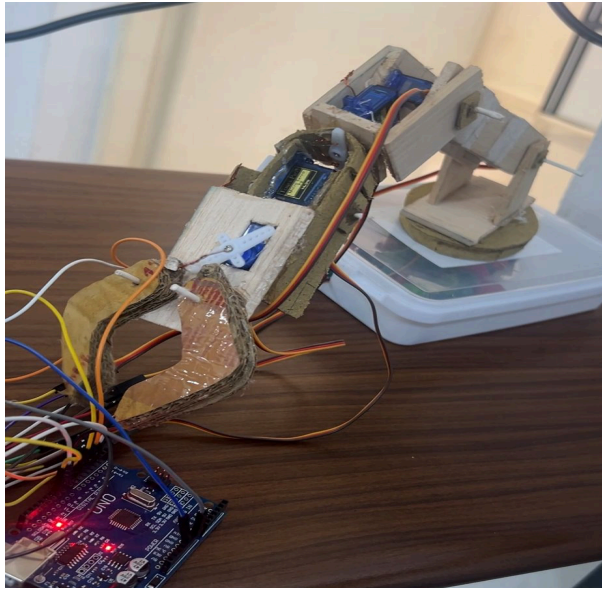


Figure 3: 3(a) Robotic manipulator in position 1, 3(b) Robotic manipulator in position 2, 3(c) Gripper open, 3(d) Gripper close

Applications

Robotic arms with multiple degrees of freedom have numerous applications, including:

- Industrial automation
- Pick-and-place operations
- Assembly line tasks
- Educational robotics experiments
- Medical robotics research
- Hazardous environment handling

Advantages

- Low-cost prototype
- Easy to modify and upgrade
- Good platform for robotics learning
- Demonstrates multi-axis motion control

Challenges

During the development of the 6-DOF robotic arm, several practical challenges were encountered. One major difficulty was maintaining mechanical stability and proper alignment of the joints, especially since lightweight materials such as cardboard or wood were used for the prototype structure. While these materials made fabrication simple and low-cost, they sometimes caused slight structural flexibility that affected the accuracy of arm movements.

Another challenge involved coordinating multiple servo motors simultaneously. Achieving smooth and synchronized motion required careful adjustment of servo angles in the Arduino program. Power supply management was also important because servo motors require higher current, and insufficient power could lead to unstable operation.

Additionally, the use of small low-torque servo motors created limitations in lifting capacity. In some cases, the motors experienced reduced performance or failure when handling heavier loads, which affected the reliability of the arm during pick-and-place tasks. Precise positioning of the gripper was also difficult without advanced control techniques such as inverse kinematics, requiring manual tuning of servo angles to achieve the desired movement.

Future Improvements

The system can be further improved by:

- Using metal or 3D-printed structures
- Adding inverse kinematics algorithms
- Integrating vision systems for object detection

- Adding wireless or joystick control
- Using higher torque servo motors

Conclusion

This project successfully demonstrates the design and implementation of a **6-DOF robotic arm using Arduino and servo motors**. The prototype provides a practical understanding of robotic manipulation, actuator control, and embedded systems. Such systems form the foundation for advanced robotic applications in automation, research, and industrial systems.